Complete-source coefficient transport and pair-spectrum bounds for the theta heat flow

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Abstract

We develop finite-population estimates for the canonical theta heat flow, retaining the complete field and its exterior contribution. The bounded pair-spectrum trace admits an exact endpoint collision value, a sharp finite multiplicity consumer, and a forest-polynomial representation in prepared coefficients. A contour equation for those coefficients is driven by commuting degree-lowering operators; at fixed degree its backward propagator is an explicit finite polynomial in integrated source moments. These statements permit quantitative error accounting without assuming internal root separation. We also construct a holomorphic correction to the effective Riemann–Siegel reference and give its field and heat-defect budgets in a microscopic real-time regime. A complete-source interval calculation on one fixed low-height box certifies its entire two-root population and discriminant throughout a complex time square. Direct evaluation is stronger for the terminal filter in this comparison. A bounded complete-cap calculation also encloses every face, but both lower bounds are negative and the direct interval is slightly narrower. The height-uniform microscopic upper estimate required for a global simplicity conclusion remains unproved; this manuscript proves neither RH nor a new global simplicity proportion.

1 Purpose, scope and normalization

The aim is to turn complete source information into quantitative control of a finite collection of zeros of the theta heat flow, and to determine which resulting bounds could support a global simplicity theorem. We separate three levels of result: finite identities with paid perturbation bounds; a bounded computation for the unchanged theta field; and the independent, still missing source estimate needed at growing height. This distinction is essential when a generic heat identity also holds for collision controls.

Throughout,

$$H(\tau,z)=\int_0^\infty e^{\tau u^2}\Phi(u)\cos(zu)\,du, \tag{1}$$
$$\Phi(u)=\sum_{n\ge1}(2\pi^2n^4e^{9u}-3\pi n^2e^{5u})e^{-\pi n^2e^{4u}}.$$

The source is positive for $u\ge0$, has superexponential decay, and defines an entire function of both variables. With the standard completed zeta function,

$$H_\tau=-H_{zz},\qquad 8H(0,z)=\xi(1/2+iz/2). \tag{2}$$

These conventions agree with [1, 2]; the real H-coordinate is twice the usual zeta ordinate.

A *complete finite population* means every spatial zero inside a declared enclosing wall, counted with multiplicity. A nonzero wall keeps this count constant, including through internal collisions. Its monic prepared polynomial P gives $H = UP$ in the interior, with U nonvanishing. The factor U, equivalently complete field data on the wall, retains the exterior contribution. None of the results below permits replacing this population by a selected subset of convenient real roots.

The main results are organized by their logical roles. The spectral and forest results give consumers of source estimates. The coefficient equation provides a finite transport mechanism. The corrected reference supplies explicit source and defect budgets, but still requires a whole-wall lower bound. The finite theta certificate shows that the interfaces can actually be supplied on one prescribed box. The physical-cap comparison retains the full boundary and its spatial complement.

Assumptions and research status. Real-spectrum positivity and the global adjacent-gap comparison are used only where explicitly stated. A local Hermite birth, a complex contour identity, or a finite complete-source certificate does not itself require RH. Conversely, excluding multiple roots does not establish reality of all roots and therefore does not prove RH. No priority claim is made for the finite algebraic identities or the combination of classical ingredients. The repository research notes give the derivation history; this draft collects selected statements and proofs into one reviewable account.

2 Collision spectra, positive-time transfer, and the missing global estimate

For distinct real points $x_1, \dots, x_N$, write

$$L(x) = \sum_{i<j} \frac{(e_i - e_j)(e_i - e_j)^T}{(x_i - x_j)^2}, \qquad q(t) = \operatorname{Tr}\Big(4tL(x(t))[I + 4tL(x(t))]^{-1}\Big).$$

A selected population is *complete at its endpoint sites* if it includes all multiplicity labels at every site it selects. Its finite matrix contains only those labels, but their motion is always that of the complete field. For endpoint multiplicities m_a, distinguish

$$M_0 = \sum_{m_a \geq 2} m_a, \qquad D_0 = \sum_a (m_a - 1), \qquad N = \sum_a m_a.$$

Thus M_0 counts labels at nonsimple sites; D_0 counts excess multiplicity.

Proposition 2.1 (Endpoint spectrum and scalar multiplicity encoding)**.** *Let a finite complete endpoint population of a real-type analytic solution of $H_t = -H_{zz}$ have real sites, with real simple descendants for small $t > 0$. Then*

$$\operatorname{spec}(4tL(x(t))) \longrightarrow \bigsqcup_a \{0, 1, \dots, m_a - 1\}, \qquad q_0 := \lim_{t \downarrow 0} q(t) = \sum_a (m_a - \mathsf{H}_{m_a}), \quad \mathsf{H}_m = \sum_{j=1}^{m} \frac{1}{j}.$$

In particular $D_0/2 \leq q_0 \leq D_0$ and $M_0 \leq 4q_0$.

Proof. The local heat expansion gives $x_{a,j}(t) = a + \sqrt{2t}\, h_{m_a,j} + O(t)$, where the $h_{m,j}$ are the zeros of the probabilists' monic Hermite polynomial He_m. The Hermite equation implies $\sum_{j \neq i} (h_i - h_j)^{-1} = h_i/2$. Acting with $L(h)$ on evaluations of the monomial x^k gives $(k/2)h_i^k$ plus lower-degree terms. The evaluations of degrees $0, \dots, m-1$ form a Vandermonde basis, so $\operatorname{spec} L(h) = \{0, 1/2, \dots, (m-1)/2\}$.

Inter-site entries, including their diagonal contributions, vanish after multiplication by t. The block limits therefore give the stated spectrum. Applying $u \mapsto u/(1+u)$ yields $\sum_{k=1}^{m-1} k/(k+1) = m - \mathsf{H}_m$. Each positive summand lies between $1/2$ and 1; also $m - \mathsf{H}_m \geq m/4$ for $m \geq 2$, with equality at $m = 2$. □

The Hermite spectrum is inherited from the local analysis in Note 06; its scalar trace is an exact encoding, not a theta upper estimate. A double contributes $1/2$, a triple $7/6$, and a quadruple $23/12$. These are fixed-population limits, with no uniform exchange of height and time limits asserted.

Theorem 2.2 (Sharp positive-time transfer)**.** *Suppose, in addition, that all adjacent descendants within each selected endpoint cluster remain globally adjacent in a complete real simple system on* $(0, T]$*, and that their gaps satisfy*

$$(g^2)' \leq 8, \qquad g(0) = 0.$$

Then $M_0 \leq 4q(t)$ *for every* $0 < t \leq T$*. The constant four is sharp.*

Proof. Every such gap obeys $g(t)^2 \leq 8t$. Partition each even cluster into adjacent pairs, and each odd cluster of size at least three into pairs and one consecutive triple. Retain only the pair edges and the two consecutive edges in each triple. Then $4tL \succeq A_*$, where the pair and triple blocks of A_* are one half of the respective path Laplacians. The map $f(A) = I - (I + A)^{-1}$ is operator monotone on positive semidefinite matrices. A pair block has spectrum $\{0, 1\}$ and contributes $1/2$ to $\operatorname{Tr} f(A_*)$. A triple has spectrum $\{0, 1/2, 3/2\}$ and contributes $1/3 + 3/5 = 14/15 \geq 3/4$. Thus each block pays at least one quarter of its labels. Singleton sites need no payment. For sharpness, $H(t, z) = z^2 - 2t$ gives $M_0 = 2$ and $q(t) = 1/2$. □

For canonical theta, the hypothesis on gaps is supplied by the full canonical real-root motion. For globally adjacent roots $a < b$, $g = b - a$,

$$(g^2)' = 8 - 4g^2 \sum_{r \neq a,b} \frac{1}{(a - r)(b - r)} \leq 8.$$

The exterior sum is part of the hypothesis and is absolutely convergent; its terms are nonnegative by global adjacency. No external factor is set to one. In the conditional theta application, RH supplies the required complete real system at positive time. The finite matrix need not contain the exterior roots, but a truncated motion law is insufficient. The improvement from the earlier constant six concerns proportional transfer; the sufficient all-simple threshold remains $q(t) < 1/2$.

Theorem 2.3 (Bounded spectral mass and short adjacent gaps)**.** *For arbitrary* $y_1 < \cdots < y_N$*, set* $A = L(y)$ *and*

$$B_N(\delta) = \frac{1}{N} \#\{i < N : y_{i+1} - y_i < \delta\}, \qquad Q_N(s) = \frac{1}{N} \operatorname{Tr}\Big(sA(I + sA)^{-1} \Big).$$

For $s, \delta > 0$*,*

$$B_N(\delta) \frac{s}{\delta^2 + 2s} \leq Q_N(s) \leq 2B_N(\delta) + \frac{\pi^2 s}{3\delta^2}. \tag{3}$$

Consequently, along any sequence of configurations and positive scales,

$$Q_N(s_N) \longrightarrow 0 \quad \Longleftrightarrow \quad B_N(a\sqrt{s_N}) \longrightarrow 0 \quad \textit{for every fixed } a > 0.$$

Proof. Mark every vertex touching a short adjacent gap. If r vertices are marked, then $r \le 2NB_N(\delta)$. The unmarked vertices are δ-separated; every marked point is also at distance at least δ from each unmarked point. Otherwise the unmarked point's nearest original neighbor on that side would be less than δ away.

Let D be the principal submatrix of A on the unmarked set. Its diagonal *retains* interactions with marked vertices. For an unmarked vertex, its kth unmarked neighbor on either side is at distance at least $k\delta$. For a marked vertex, its kth unmarked neighbor on either side satisfies the same bound. Summing $2\sum_{k\ge1}(k\delta)^{-2}$ first over the unmarked vertices and then over marked-to-unmarked interactions gives

$$\operatorname{Tr} D \le \frac{\pi^2 N}{3\delta^2}.$$

Interlacing pays at most r for the discarded eigenvalues; the remaining filter values are bounded by s times the eigenvalues of D. This proves the upper bound, including the empty-principal-matrix case.

The short adjacent edges form disjoint paths. From K such edges select a matching of at least $K/2$ edges. Each retained edge has a Laplacian eigenvalue at least $2/\delta^2$. Dropping all other edges and using operator monotonicity proves the lower bound. Substituting $\delta = a\sqrt{s}$ gives

$$B_N(a\sqrt{s}) \le (a^2+2)Q_N(s), \qquad Q_N(s) \le 2B_N(a\sqrt{s}) + \frac{\pi^2}{3a^2}.$$

The equivalence follows by taking a limsup at fixed a, then letting $a \to \infty$ for the reverse direction. □

For example, independently acquired bounds $B_N(\delta) \le C\delta^\beta + \varepsilon_N$, $\beta > 0$, imply

$$Q_N(s) \le \left(2C + \frac{\pi^2}{3}\right) s^{\beta/(\beta+2)} + 2\varepsilon_N$$

whenever $s^{1/(\beta+2)}$ lies in their certified range. This is a sufficient source input, not a bound proved here for theta.

To state the global interface precisely, let an endpoint target have N_h labels, of which $n_h = N_h - r_h$ are covered by complete clusters. At a source-prescribed time t_h, set $y_i = \nu_h x_i(t_h)$ and $s_h = 4t_h\nu_h^2$. Under Theorem 2.2,

$$\frac{M_h}{N_h} \le \frac{r_h}{N_h} + 4\frac{n_h}{N_h} Q_{n_h}(s_h). \tag{4}$$

Thus density-one simplicity would follow conditionally from $r_h = o(N_h)$ and a vanishing fraction of gaps smaller than $2a\sqrt{t_h}$ for every fixed $a > 0$. That microscopic gap estimate remains unpaid. Choosing times after resolving the unknown endpoint gaps would merely reformulate the target. The published late-clock onset $x \ge e^{C/t}$ in [1] does not supply this estimate. With its natural unfolding $\nu_X = g_x(X,t) \sim \log X/(4\pi)$, the uniformly bounded normalized gaps instead imply $Q_N(s) \ge 1 - 1/N - (1 + 4s/C_g^2)^{-1/2}$ by path comparison. For this unfolding, $s = 4t\nu_X^2 \to \infty$ in the high-height onset regime, so the unmodified filter saturates. This does not exclude matched backward transport on a different clock.

3 A coefficient-level filter with explicit conditioning

The preceding filter is defined through roots, but its acquisition need not isolate them individually. Let $P(z) = \prod_{i=1}^N (z - x_i)$ be monic and $\Delta = \prod_{i<j}(x_i - x_j)^2$. The following identity is algebraic; real rootedness is needed only for the positivity conclusions.

Proposition 3.1 (Forest polynomial and coefficient enclosure)**.** *For $a > 0$ and distinct roots define $D = \Delta \det(I + aL)$ and $G = a\partial_a D$. Then*

$$D = \sum_{F \text{ forest}} a^{|E(F)|} \prod_{C \in \mathrm{comp}(F)} |C| \prod_{\{i,j\} \notin E(F)} (x_i - x_j)^2, \qquad q_a = \frac{G}{D}. \tag{5}$$

The sum runs over all spanning labeled forests, including isolated vertices. The quotient identity is asserted where $D \neq 0$. Both D and G are symmetric polynomials in the roots and hence polynomials in the coefficients of P. Fix $q_* = G_*/D_*$ and form the coupled polynomial $K = G - q_* D$. If coefficient data give $|D - D_*| \leq E_D < |D_*|$ and $|K| \leq E_K$, then

$$|q_a - q_*| \leq \frac{E_K}{|D_*| - E_D}. \tag{6}$$

The separate allowance $E_K = \epsilon_G + |q_*|\epsilon_D$, obtained from $|G - G_*| \leq \epsilon_G$, is a valid fallback.

Proof. Apply Cauchy–Binet to the weighted incidence matrix in L. A cyclic edge selection has dependent columns and contributes zero. For a forest, its incidence Gram determinant is $\prod_C |C|$: its nonzero maximal minors omit one vertex per component and have square one. Thus its determinant contribution is the edge-weight product times $a^{|E|} \prod_C |C|$. Each edge occurs at most once, so multiplying by the single product Δ cancels every denominator. This proves the polynomial identity. Logarithmic differentiation of the determinant gives

$$a\partial_a \log D = a \operatorname{Tr}\big(L(I + aL)^{-1}\big) = q_a.$$

The discriminant is independent of a. Finally $q_a - q_* = K/D$, and $|D| \geq |D_*| - E_D$, proving (6). □

The coefficient residuals in the proposition are finite, computable quantities. For example, on a coefficient polydisc $|c_j - c_{*,j}| \leq r_j$, take $D_* = D(c_*)$, $G_* = G(c_*)$ and use the finite Taylor majorants $E_D = \sum_{\alpha \neq 0} |\partial^\alpha D(c_*)| r^\alpha/\alpha!$ and $E_K = \sum_{\alpha \neq 0} |\partial^\alpha K(c_*)| r^\alpha/\alpha!$, since $K(c_*) = 0$. Form K before taking absolute values, so correlated coefficient changes can cancel. The estimating gain is that the paid margin is for D, not necessarily for Δ or individual roots. It is not a statement that all coefficient balls are well conditioned.

Lemma 3.2 (Real repeated-root boundary)**.** *For fixed $a > 0$ and a real root multiset, the polynomial extension has $D > 0$ if and only if every root multiplicity is at most two.*

Proof. All forest terms are nonnegative. A nonzero term must include every edge joining equal roots, since any omitted such edge contributes zero. A cluster of three or more forces a cycle and makes every term vanish. If all clusters have size at most two, the disjoint collapsed pair edges form a forest whose complementary gaps are nonzero, so its term is positive. □

The distinction is visible already in degrees two and three. For $P = z^2 + bz + c$, put $\delta = b^2 - 4c$. Then

$$D = \delta + 2a, \qquad G = 2a, \qquad q_a = \frac{2a}{\delta + 2a}.$$

In particular, $P = z^2 + p$, $-a/8 \leq p \leq 0$, gives $4/5 \leq q_a \leq 1$, including the repeated reference polynomial. There is no discriminant floor or internal root isolation in this enclosure. It is a stable filter calculation, not a simplicity certificate: its values are above the relevant threshold. For complex $|p| \leq a/8$, one similarly has $|D| \geq 3a/2$ and $|q_a - 1| \leq 1/3$.

For $P = z^3 + bz^2 + cz + d$, let $B = b^2 - 3c$. Directly from the three-vertex forest sum,

$$D = \Delta + 2aB^2 + 6a^2B, \qquad G = 2aB^2 + 12a^2B.$$

On the real-rooted cone, $0 \le \Delta \le 4B^3/27$. Thus, for $B > 0$ and $\kappa = B/a$,

$$\frac{2\kappa + 12}{4\kappa^2/27 + 2\kappa + 6} \le q_a \le \frac{2\kappa + 12}{2\kappa + 6}.$$

The restricted real limit at a triple is two. There is no corresponding universal extension in complex coefficient space: along $P = z^3 - r^3$, $B = 0$, $\Delta = -27r^6$, and $q_a = 0$, whereas the real Hermite approach gives $q_a \to 2$ at fixed a.

The heat scale must also be retained. For the cubic heat polynomial $P_t = z^3 - 6tz$,

$$D(t, a) = 108t(a + 2t)(a + 4t), \qquad D(t, 4t) = 5184t^3, \qquad G(t, 4t) = 6048t^3,$$

so $q(t) = 7/6$. More generally, the coefficient snapshots

$$P_t = z^3 - 6tz + \eta t^{3/2}, \qquad q(t) = \frac{6048}{5184 - 27\eta^2}.$$

For $\eta \neq 0$ these snapshots do not form a heat solution. For real η with $|\eta| \le \sqrt{32}$ the polynomials are real rooted and $7/6 \le q \le 7/5$. A constant coefficient perturbation of order $t^{3/2}$ therefore has an order-one effect on the limiting filter. To recover the Hermite value, sufficient depressed-coefficient accuracies are $p + 6t = o(t)$ and $d_{\text{depressed}} = o(t^{3/2})$; the cost of centering must be included. Likewise the quadratic heat polynomial retains the scale ϵ/t for its constant coefficient error. Exact coefficient transport removes rootwise inverse-gap factors, but cannot remove these source precision requirements. Notes 10 and 11 give further controls and the associated complete finite replays.

4 Complete-source coefficient transport on a fixed wall

The root velocity equations introduce inverse internal gaps. Passing to the complete prepared polynomial removes those denominators from the transport equation. The source information is still required, but it can be acquired on one exterior wall. We give the finite theorem with its source cost explicit.

Fix a circle $\Gamma = \{|z - z_0| = R\}$, positively oriented, and a real time interval $[t, T]$. Suppose H is holomorphic on a neighborhood of the closed spatial disk and the time interval, satisfies

$$\partial_t H = -\partial_z^2 H, \qquad H(u, w) \neq 0 \quad (u \in [t, T],\ w \in \Gamma), \tag{7}$$

and has N zeros inside Γ, counted with multiplicity. Write $H = UP$, where P is the complete monic polynomial of those zeros and U is holomorphic and nonzero in the disk. The center and radius remain fixed. For a polynomial f, set

$$Q_f(w, z) = \frac{f(w) - f(z)}{w - z}, \qquad Z_H = \frac{H_{zz}}{H}, \qquad \mathcal{A}_H f = \frac{1}{2\pi i} \int_\Gamma Z_H(w) Q_f(w, z)\, dw.$$

The divided difference Q_f is a polynomial, including when $w = z$.

Proposition 4.1 (Complete-source equation)**.** *The prepared polynomial satisfies*

$$P_t = -\mathcal{A}_H P = -P_{zz} - 2\mathcal{T}_P(bP_z), \qquad b = U_z/U, \tag{8}$$

where

$$\mathcal{T}_P g(z) = \frac{1}{2\pi i} \int_\Gamma \frac{g(w)}{P(w)} Q_P(w,z)\, dw.$$

These identities remain valid at internal multiple zeros.

Proof. The operator $\mathcal{T}_P$ gives the degree less than N analytic remainder modulo P; in particular it fixes every polynomial of degree less than N and annihilates P times a function holomorphic in the disk. These statements follow directly from Cauchy's formula and do not require simple roots. Since $H_t/H = P_t/P + U_t/U$, multiplying by Q_P and integrating gives $P_t = \int_\Gamma (H_t/H) Q_P/(2\pi i)$. The heat equation proves the first identity. Finally $H_{zz}/U = P_{zz} + 2bP_z + (U_{zz}/U)P$, and analytic remainder gives the second. □

Put $\zeta = z - z_0$, and let S be the plain coefficient shift: $S\zeta^j = \zeta^{j-1}$ for $j \geq 1$, and $S1 = 0$. It is not d/dz. Define the complete source moments

$$M_k(t) = \frac{1}{2\pi i} \int_\Gamma (w - z_0)^k Z_H(t,w)\, dw.$$

Expanding a divided monomial gives, on polynomials of degree at most n,

$$\mathcal{A}_H = \sum_{j=1}^{n} M_{j-1} S^j. \tag{9}$$

Thus all these operators commute, including operators constructed from different source fields. They lower degree and are nilpotent.

Theorem 4.2 (Finite coefficient propagator)**.** *Let $m_j(t,s) = \int_t^s M_{j-1}(u)\, du$. The backward propagator of $E' = -\mathcal{A}_H E$ on degree at most n is*

$$\mathcal{U}_H(t,s) = \exp\left(\sum_{j=1}^{n} m_j(t,s) S^j \right). \tag{10}$$

The exponential terminates after degree n. For the fixed coefficient norm $\|f\|_R = \sum_j |f_j| R^j$, its exact induced norm is

$$B_n(t,s) = \sum_{k=0}^{n} \left| [v^k] \exp\left(\sum_{j=1}^{n} \frac{m_j(t,s)}{R^j} v^j \right) \right|. \tag{11}$$

If $V = \int_t^s \sup_\Gamma |Z_H(u,w)|\, du$, then

$$B_n(t,s) \leq \sum_{\ell=0}^{n} \binom{n}{\ell} \frac{V^\ell}{\ell!}. \tag{12}$$

Proof. Equation (9) makes the time generators commuting polynomials in S, so differentiating the finite exponential proves (10). If $\mathcal{U} = \sum b_k S^k$, the weighted column sum of its jth column is $\sum_{k=0}^{j} |b_k| R^{-k}$, maximal at $j = n$. This proves the exact norm formula. The wall estimate gives $|m_j|/R^j \leq V$. Majorize the generating series by $\exp(Vv/(1-v))$ through degree n. Summing those coefficients and using $\sum_{k=\ell}^{n} \binom{k-1}{\ell-1} = \binom{n}{\ell}$ proves (12). □

The estimate is polynomial in acquired integrated source sizes at fixed degree. It is not asserted to be polynomial uniformly in height, population size, or a late-time parameter. For a difference of monic degree-N polynomials, use $n = N - 1$ in the error propagator, while propagating the polynomial itself uses $n = N$.

4.1 Reference errors and what must be acquired

Let G be a reference holomorphic field with the same zero count and with $G(u,w)e0$ for every $u \in [t,T]$ and $w \in \Gamma$, with complete monic prepared polynomial P_*. It need not be an exact heat solution. Define

$$d_* = \partial_t P_* + \mathcal{A}_G P_*, \qquad E = P - P_*, \qquad r = (\mathcal{A}_H - \mathcal{A}_G)P_* + d_*.$$

Then $E' = -\mathcal{A}_H E - r$, and hence

$$E(t) = \mathcal{U}_H(t,T)E(T) + \int_t^T \mathcal{U}_H(t,s)r(s)\,ds. \tag{13}$$

The complete reference defect is also a wall quantity:

$$d_*(z) = \frac{1}{2\pi i}\int_\Gamma \frac{G_t + G_{zz}}{G}(w) Q_{P_*}(w,z)\,dw.$$

If $\eta_Z \geq \sup_\Gamma |Z_H - Z_G|$ and $W_R(P_*) = \sum_j j|p_{*,j}|R^j$, a closed finite error estimate is

$$\|E(t)\|_R \leq B_{N-1}(t,T)\|E(T)\|_R + \int_t^T B_{N-1}(t,s)\big(\eta_Z(s)W_R(P_*(s)) + \|d_*(s)\|_R\big)\,ds. \tag{14}$$

A directly evaluated correlated residual can replace this absolute majorant. Its size must be acquired independently; an unknown coefficient difference inside a forcing bound is not free data.

For example, acquired field and second-derivative errors e_0, e_2, reference wall floor $m_* > e_0$, and reference second-derivative upper bound $M_{2,*}$ give

$$\eta_Z \leq \frac{e_2}{m_* - e_0} + \frac{M_{2,*}e_0}{m_*(m_* - e_0)}. \tag{15}$$

An analytic field-error bound on a spatial collar of width d gives $e_2 \leq 2e_{\mathrm{collar}}/d^2$. This Cauchy step concerns the field error, not a logarithmic ratio continued through different interior zero sets.

For exact heat fields the integrated moments have the endpoint representation

$$m_j(t,T) = -\frac{1}{2\pi i}\int_\Gamma (w - z_0)^{j-1} \log\frac{H(T,w)}{H(t,w)}\,dw, \tag{16}$$

using the logarithm lifted continuously from time t. Thus equally informed endpoint wall data already determine the homogeneous propagator. The commutation theorem does not confer an intrinsic advantage on a curved path; a gain would have to come from improved source estimates or conditioning along that path, with its entire error paid.

Remark 4.3. The fixed-source generator is different from free polynomial heat evolution on arbitrary error vectors. For $H = z^3 - 6tz$, one finds $\mathcal{A}_H = 6S^2$ on degree at most three, while $\partial_z^2(z^2) = 2$. Comparing two independently evolving polynomial heat fields includes the source-quotient difference in (13). Conversely, suppressing the external unit in (8) changes the complete problem. Note 12 contains the free-heat comparison, complex-path formulation and collision controls.

5 A complete reference at microscopic real heat times

We now specify a theta source input rather than assume a small wall error. Throughout,

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad H_0(z) = \tfrac{1}{8}\xi(\tfrac{1}{2} + \tfrac{iz}{2}).$$

The effective approximation of Polymath [1] applies for real $0 < t \le 1/2$, $x \ge 200$, $0 \le y \le 1$. In this section $z = x + iy$ and

$$N = \left\lfloor \sqrt{x/(4\pi) + t/16} \right\rfloor.$$

We work on a chart where N stays fixed. Changes of N require explicit patching; a jumping truncation is not a holomorphic reference.

Define, with principal logarithms away from their cuts,

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}\exp[(s/2 - 1/2)\operatorname{Log}(s/2) - s/2], \quad \alpha(s) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2}\operatorname{Log}\frac{s}{2\pi}.$$

Here $\alpha = M_0'/M_0$. Put $s = (1 - iz)/2$ and

$$G_{t,N}(z) = \sum_{\sigma\in\{s,1-s\}}\sum_{n=1}^{N} M_0(\sigma)n^{-\sigma}\exp\left[\frac{t}{4}(\alpha(\sigma) - \log n)^2\right], \qquad B_t(z) = M_0(s)e^{t\alpha(s)^2/4}. \tag{17}$$

This rewrites the complete two-sum reference without complex conjugates. In particular G is locally holomorphic in t, z, although it is not an exact heat solution.

5.1 Explicit field and heat-defect budgets

For completeness, a finite coarsening of the imported error bounds is as follows. Set $X = x/(4\pi)$, $L = \log X$, and

$$p = \operatorname{Re}(s + t\alpha(s)/2), \quad \rho = e^{0.02y}X^{-y/2}, \quad \nu = \frac{ty}{2(x-6)},$$

$$S_0 = \sum_{n=1}^{N}(1 + \rho N^\nu n^y)e^{t\log^2 n/4}n^{-p}, \quad b_0 = \frac{t^2L^2/16 + 0.626}{x - 6.66}, \quad E_{AB} = S_0(e^{b_0} - 1).$$

The exact value of p avoids relying on slightly different printed lower-bound corrections for its real part. Theorem 1.3 of [1] gives $|H_t - G_{t,N}|/|B_t| \le E_{AB} + E_{C,0}$, with

$$E_{C,0} = X^{-(1+y)/4}\exp\left[-\frac{tL^2}{16} + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{3|L + i\pi/2| + 10.44}{x - 12}\right].$$

For $tL^2 \le K$ and $y = 0$, $E_{AB} = O_K(X^{-3/4})$, whereas $E_{C,0} = O_K(X^{-1/4})$. Indeed the retained weights have modulus at most a constant times $n^{-1/2}$, and their relative error bracket is $O_K(1/x)$. These are usable absolute errors even when $tL^2 \to 0$; the late-clock assumption $x \ge e^{C/t}$ is not needed. That assumption is needed for the much shorter two-carrier asymptotic, whose omitted absolute sum is not small on the microscopic clock.

For a term $g = M_0(\sigma)n^{-\sigma}e^{t\beta^2/4}$, with $\beta = \alpha(\sigma) - \log n$, direct differentiation gives

$$(\partial_t + \partial_z^2)g = -\frac{g}{4}\left[\alpha' + t\beta^2\alpha' + \frac{t}{2}((\alpha')^2 + \beta\alpha'') + \frac{t^2}{4}\beta^2(\alpha')^2\right]. \tag{18}$$

Here both $\sigma = s$ and $\sigma = 1 - s$ satisfy $\partial_z^2 = -\partial_\sigma^2/4$. Taking absolute values in this finite sum is an explicit defect budget. Since $\beta = O(\log x)$, $\alpha' = O(x^{-1})$, $\alpha'' = O(x^{-2})$, it proves

$$|G_t + G_{zz}|/|B_t| = O_K(X^{-3/4}) \qquad (y = 0,\ tL^2 \le K). \tag{19}$$

No source quotient has been divided by an unacquired wall floor in this step.

5.2 An analytic continuation of the explicit correction

Corollary 6.4 and Proposition 6.6(vi) of [1] improve the field remainder by retaining a correction C_t. Define

$$v_0 = x/2 + \pi t/8, \quad p_N(v) = 1 + 2N - \sqrt{2v/\pi},$$

$$U(v) = \exp\left[-i\left(\frac{v}{2}\log\frac{v}{2\pi} - \frac{v}{2} - \frac{\pi}{8}\right)\right], \quad C_0(p) = \frac{e^{\pi i(p^2/2+3/8)} - i\sqrt{2}\cos(\pi p/2)}{2\cos(\pi p)}.$$

The singularities at $p = \pm 1/2$ are removable, with value $(1-i)/4$. The displayed source correction is

$$C_t(x+iy) = 2(-1)^N e^{-\pi i y/8 + t\pi^2/64}\,\mathrm{Re}\{M_0(iv_0)C_0(p_N(v_0))U(v_0)e^{\pi i/8}\}.$$

It gives

$$\frac{|H_t - (G_{t,N} - C_t)|}{|B_t|} \le E_{AB} + E_C, \tag{20}$$

where

$$E_C = X^{-(1+y)/4} e^{-tL^2/16 + (3|L+i\pi/2|+3.58)/(x-8.52)}\left[\frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{6.92}{x - 12}\right].$$

The extra $1/N$ makes this remainder $O_K(X^{-3/4})$ on the real axis. However, the displayed C_t is generally not holomorphic in z; its real-part operation cannot be differentiated as an analytic reference.

There is an explicit analytic replacement. Let $C_0^{\#}(p) = \overline{C_0(\bar{p})}$, $C_R = (C_0 + C_0^{\#})/2$, $C_I = (C_0 - C_0^{\#})/(2i)$, and

$$V_N(v) = v^{3/2} C_R(p_N(v)) - v^{1/2} C_I(p_N(v)), \qquad k_N = -(-1)^N \frac{\sqrt{\pi}}{4}.$$

The functions C_R, C_I are holomorphic continuations of the real and imaginary parts on the real axis, not real-part operations on complex arguments. Set

$$\widetilde{C}_{t,N}(z) = k_N e^{-\pi z/8 - \pi^2 t/64} V_N(z/2 + \pi t/8), \qquad \widehat{G} = G - \widetilde{C}. \tag{21}$$

Use principal square roots on $\mathrm{Re}\, v > 0$.

Proposition 5.1 (Corrected analytic reference)**.** *The correction* (21) *agrees with C_t at $y = 0$. With*

$$Q(t,z) = \frac{\sqrt{\pi}}{4}\frac{e^{-\pi x/8 - \pi^2 t/64}}{|B_t(z)|},$$

it satisfies the explicit bounds

$$\frac{|H_t - \widehat{G}|}{|B_t|} \le E_{AB} + E_C + Q(t,z)\frac{|y|}{2}\sup_{0\le\theta\le 1}|V_N'(v_0 + i\theta y/2)|, \tag{22}$$

$$(\partial_t + \partial_z^2)\widetilde{C} = \frac{k_N}{4} e^{-\pi z/8 - \pi^2 t/64} V_N''(z/2 + \pi t/8). \tag{23}$$

For fixed K and $y_0 < 1$, its normalized field allowance is $O_{K,y_0}(X^{-(3+y)/4})$ uniformly for $0 \le y \le y_0$ and $0 < tL^2 \le K$. Its correction defect is $O_{K,y_0}(X^{-(5+y)/4})$. In particular both the field error and total heat defect of $\widehat{G}$ are $O_K(X^{-3/4})$ at $y = 0$.

Proof. For positive real v, substituting $\operatorname{Log}(iv/2) = \log(v/2) + i\pi/2$ gives the exact cancellation

$$M_0(iv)U(v)e^{\pi i/8} = -\frac{\sqrt{\pi}}{8}\sqrt{v}\,(v+i)e^{-\pi v/4}.$$

Consequently $C_t(x+iy) = k_N e^{-\pi(x+iy)/8-\pi^2 t/64}V_N(v_0)$. The fundamental theorem of calculus from v_0 to $v_0 + iy/2$ proves (22). In the heat defect, the coefficients of V_N and V_N' cancel because the argument derivatives are $1/2$ and $\pi/8$, while the exponential logarithmic derivatives are $-\pi/8$ and $-\pi^2/64$. This proves (23).

For completeness the derivative sizes have explicit, zero-independent constants. Along the comparison segment,

$$|p_N(v) - p_N(v_0)| \le \frac{|y|}{2\sqrt{2\pi v_0}} < \frac{1}{8},$$

so $|p_N(v)| < 9/8$. On this disk both $C_R^{(j)}$ and $C_I^{(j)}$ are bounded by $K_j = j!8^j M_*$ for $j = 0, 1, 2$, where

$$M_* = \frac{e^{25\pi/32} + \sqrt{2}\cosh(5\pi/8)}{2\sinh(\pi/8)}.$$

Indeed $|\cos \pi p| \ge \sinh(\pi/8)$ on $|p| = 5/4$: split into $|\operatorname{Im} p| \ge 1/8$ and its complement, where $1 < |\operatorname{Re} p| \le 5/4$. Bound the numerator on that circle, remove $\pm 1/2$, and apply Cauchy's estimate with radius $1/8$. Since $p_N' = -1/\sqrt{2\pi v}$ and $p_N'' = -p_N'/(2v)$, differentiation gives $V_N' = O(|v|)$ and $V_N'' = O(|v|^{1/2})$ with these explicit constants. Finally $Q = O_{K,y_0}(X^{-(7+y)/4})$ follows directly from the displayed Stirling factor. Combine these estimates with (18). Note 14 records the full derivative formulas and convenient finite majorants. □

5.3 The remaining wall and time gates

These are absolute source allowances. To use the coefficient theorem, acquire a wall floor $m_f > e_0$ for $f = \widehat{G}/B$, where e_0 bounds the right-hand side of (22). Then $|H/\widehat{G} - 1| \le e_0/m_f$. For a common analytic normalization write $e = H/B - f$ and $\lambda = B'/B$. The complete quotient cancellation is

$$\frac{H_{zz}}{H} - \frac{\widehat{G}_{zz}}{\widehat{G}} = \frac{fe'' - ef'' + 2\lambda(fe' - ef')}{f(f+e)}.$$

Thus an acquired spatial collar supplies derivative errors without introducing internal root gaps. The local reference derivative ratios still require bounds. A collar crossing the real axis must convert the reflected error to one analytic carrier before applying Cauchy's estimate.

All imported field errors here concern real t. The explicit references extend analytically in complex time on their charts, but that does not extend the error for H with the same bound. A complex-time route needs a separately paid source remainder. Finally a fixed-N chart is not the same as a fixed number of roots in a selected block: a prepared population count, common wall, and the forest denominator remain separate acquisition obligations. The finite certificate in Note 13 pays those obligations in its original bounded box. No estimate in this section supplies the missing unbounded-height small-gap upper bound or a global simplicity conclusion.

6 Complete-source acquisition on a fixed theta box

The following calculation supplies actual canonical source data to the coefficient interface. Its purpose is a bounded calibration, not a new high-height zero result. The parameters were fixed before acquisition:

$$\Omega = [26, 46] + i[-2, 2], \qquad D = [-10^{-4}, 10^{-4}] + i[-10^{-4}, 10^{-4}]. \tag{24}$$

Every zero in the rectangle is included. No individual root location, root disk, RH assumption or internal gap is an input.

Theorem 6.1 (Finite complete-theta certificate)**.** *For the unchanged field* (1) *on* (24)*, the interval computation and analytic error bounds below establish*

$$\inf_{D\times\partial\Omega} |H| > 4.94 \cdot 10^{-7}, \qquad \inf_{D\times\Omega} (|H|^2 + |H_z|^2) > 1.1178 \cdot 10^{-13}.$$

The complete spatial population has multiplicity count two for every $\tau \in D$. Its centered monic quadratic has discriminant satisfying

$$189.68 < \Re\Delta(\tau) < 189.80, \qquad |\Im\Delta(\tau)| < 0.058.$$

Hence both roots are distinct throughout D, and real and simple for every real $t \in [-10^{-4}, 10^{-4}]$. At $t = 10^{-4}$,

$$4.2149 \cdot 10^{-6} < q(t) < 4.2176 \cdot 10^{-6} < \tfrac{1}{2}.$$

These conclusions concern only this finite population.

6.1 Uniform error for the complete source

Arb ball arithmetic [3], through python-flint 0.9.0 at 192-bit precision, encloses the first 96 Taylor coefficients of $H_0(z) = \xi(1/2 + iz/2)/8$ about the real center 36. Real-type symmetry justifies real coefficients; the computed imaginary coefficient balls are checked to contain zero. Denote the degree-95 polynomial by P_{95}. Positivity of Φ gives

$$|H_0(36 + w)| \le H_0(24i) \quad (|w| \le 24),$$

so the full spatial Taylor remainder on the radius-12 disk is at most

$$E_{\mathrm{sp}} = 2H_0(24i)2^{-96}. \tag{25}$$

On the radius-11 disk, Cauchy's inequality on a radius-one circle gives the derivative error $j!E_{\mathrm{sp}}$. The physical rectangle lies strictly inside that disk since $\sqrt{104} < 11$.

Put $T_* = 1/5000$, so the entire time square, including its outward interval representation, lies in $|\tau| < T_*$. Use

$$G_\tau(z) = \sum_{k=0}^{5} \frac{(-\tau)^k}{k!} P_{95}^{(2k)}(z). \tag{26}$$

For $|\Im z| \le 2$, Taylor's integral remainder for the source exponential and (25) give

$$|H_j(\tau, z) - G_\tau^{(j)}(z)| \le E_{\mathrm{sp}} \sum_{k=0}^{5} \frac{T_*^k (j + 2k)!}{k!} + \frac{T_*^6}{6!} M_{12+j}, \qquad 0 \le j \le 3, \tag{27}$$

where

$$M_j = \int_0^\infty u^j e^{T_* u^2} \Phi(u) \cosh(2u)\, du.$$

Every source index and the entire integration tail are paid. In fact, with

$$C_\Phi = \frac{2\pi^2}{1 - 16e^{-3\pi}}, \quad B_j = T_*/e^2 + (11 + j)/(2e), \quad X_* = (\pi/2)e^4,$$

one has

$$M_j \le e^{T_*} H_0(2i) + \frac{C_\Phi}{4X_*} \exp\{B_j^2/(2\pi) - X_*\}. \tag{28}$$

For $u \le 1$, retain $\cosh(2u)$ and bound $u^j e^{T_* u^2}$ by e^{T_*}. For $u \ge 1$, use $\Phi \le C_\Phi e^{9u - \pi e^{4u}}$, $u^j \le e^{ju}$, $u^2 \le e^{2u}/e^2$, and $u \le e^{2u}/(2e)$. Completing a square in e^{2u}, followed by $\Gamma(0, X_*) \le e^{-X_*}/X_*$, proves (28). All four errors in (27) are below $6 \cdot 10^{-27}$. This is an approximation to the complete field, not a replacement source.

6.2 Wall count and coefficient enclosure

On each accepted wall segment γ_ν, the source oracle gives

$$m_\nu \le \inf |H_0|, \qquad \omega_\nu \ge \sup_{D \times \gamma_\nu} |H_\tau - H_0|, \qquad \eta_\nu = \omega_\nu / m_\nu < 1/4.$$

The time difference is evaluated with its constant heat term cancelled before interval evaluation. Sixteen segments, found in 28 cell calls, cover the complete closed rectangle. The largest η_ν is less than $3.82 \cdot 10^{-4}$. Thus Rouché's theorem transfers the entire multiplicity count from the anchor.

At the anchor, validated integration encloses

$$p_k(0) = \frac{1}{2\pi i} \oint_{\partial\Omega} (w - 36)^k \frac{H_0'(w)}{H_0(w)}\, dw, \qquad k = 0, 1, 2. \tag{29}$$

The meromorphic quotient formed from the entire finite polynomial is passed to the validated integrator. It is holomorphic near the certified wall; the integrator checks its required complex neighborhoods. A physical-cell range enclosure, which is valid only on its declared physical subset, is not an analytic callback. If $|H_0| \ge m$, $|H_0'| \le L$, and source errors are e_0, e_1, the external correction is

$$\left| \frac{H_0'}{H_0} - \frac{G_0'}{G_0} \right| \le \frac{e_1}{m} + \frac{(L + e_1)e_0}{m(m - e_0)}, \qquad m > e_0. \tag{30}$$

Here $e_0 = e_1 = E_{\mathrm{sp}}$. Integrating this bound on the accepted segments pays the complete-source correction. The resulting p_0 enclosure isolates the unique integer two. The argument principle therefore supplies the count rather than a numerical guess about root locations.

The principal relative logarithm $\ell_\tau = \mathrm{Log}(H_\tau / H_0)$ is single-valued around the wall and obeys $|\ell_\tau| \le -\log(1 - \eta_\nu)$. Integration by parts gives

$$\begin{aligned} p_k(\tau) - p_k(0) &= -\frac{k}{2\pi i} \oint (w - 36)^{k-1} \ell_\tau(w)\, dw, \\ |p_k(\tau) - p_k(0)| &\le \frac{k}{2\pi} \sum_\nu |\gamma_\nu| r_\nu^{k-1} [-\log(1 - \eta_\nu)], \end{aligned} \tag{31}$$

where $r_\nu = \sup_{\gamma_\nu} |w - 36|$. No internal gap or interior logarithmic ratio is differentiated. The acquired values are

$$p_1(0) = -1.6864704389875024 \pm 7.4 \cdot 10^{-12}, \quad p_2(0) = 96.29229323297757 \pm 4.1 \cdot 10^{-11},$$

with these bounds on each complex component. The uniform continuation errors are below 0.001772 and 0.025736, respectively. Since $\Delta = 2p_2 - p_1^2$, outward interval arithmetic proves the discriminant enclosure in Theorem 6.1. Conjugation symmetry gives real coefficients on real time slices; positivity of the discriminant then supplies reality and simplicity. The quadratic forest formula gives $q = 8t/(\Delta + 8t)$.

For the first-jet floor, 31 closed spatial rectangles, obtained in 61 cell calls, cover Ω while every cell uses all of D. The sum of the squared lower bounds on $|H|$ and $|H_z|$ is positive on every cell. Exact rational area, containment and disjointness checks verify the cover. This is also a direct quantitative certificate of regularity on every physical cap and face.

6.3 Matched direct comparison and reproducibility

Direct evaluation of (29) at $t = 10^{-4}$, using the same source, wall, tolerance and correction, gives the much tighter ball

$$q(10^{-4}) = 4.2162799666292433 \cdot 10^{-6} \ \pm \ 2.34 \cdot 10^{-18}.$$

The uniform complex-time transport ball has radius $1.28 \cdot 10^{-9}$. The different information delivered by the two calculations should be kept in view: the latter controls all of D, whereas the former is a direct terminal enclosure. For real-time simplicity alone, four source values give sign changes in $(28, 29)$ and $(42, 43)$, uniformly on the real interval; together with the shared complete count they give a cheaper direct conclusion. No practical advantage for continuation is established here.

Research Note 13 and Review 11 contain the detailed proof and audit. The sibling `numerics` folder contains the source core, the `07` source and wall scripts and their small JSON receipts, and the `08` complete first-jet cover. The installed wall replay passes 109 assertions, the independent complete-source quadrature checks pass 46, and the full-volume cover passes five. The source checks include four direct theta-integral evaluations with both omitted modes and the integration tail paid. They supplement the uniform analytic bounds, rather than proving them by sampling. The receipts record numerical precision, source errors, geometry, script hashes and timings. Typical wall work is a fraction of a second; the anchor and direct contour integrations take several seconds on the recorded machine. These timings are implementation observations, not asymptotic complexity claims.

7 Complete physical contours and matched comparisons

A contour calculation must preserve the observable before it estimates its pieces. Write $\tau = t + i\sigma$, $z = x + iy$, and $H_j = \partial_z^j H$ for a holomorphic solution of $H_\tau = -H_{zz}$. For the first jet $G = (H_0, H_1)$, put

$$E = |H_0|^2 + |H_1|^2, \qquad L = H_1^2 - H_0 H_2, \quad N = H_2^2 - H_1 H_3, \quad M = H_1 H_2 - H_0 H_3.$$

Where $E > 0$, the normalized sphere-volume form is

$$\mathcal{C} = \frac{\mathcal{A} \wedge d\mathcal{A}}{4\pi^2}, \qquad \mathcal{A} = \frac{\mathrm{Im}(G^* dG)}{E}.$$

Define

$$(f, p) = \frac{(\Re(L\overline{N}), \Im(L\overline{N}))}{2\pi^2 E^2}, \qquad (r, s) = \frac{(\Re(M\overline{N}), \Im(M\overline{N}))}{2\pi^2 E^2}.$$

Direct evaluation on the coordinate vectors, using $G_t = (-H_2, -H_3)$, $G_\sigma = iG_t$, $G_x = (H_1, H_2)$ and $G_y = iG_x$, gives

$$\mathcal{C} = f\, d\sigma \wedge dx \wedge dy - p\, dt \wedge dx \wedge dy + r\, dt \wedge d\sigma \wedge dy - s\, dt \wedge d\sigma \wedge dx. \tag{32}$$

These formulas use no projective denominator such as H or H_z alone. A zero of either coordinate is harmless when $E > 0$.

Let $\Omega = [A, B] + i[-b, b]$ and $W_- = [-\epsilon, 0] \times [-Y, Y] \times \Omega$. Give the cap at fixed t the forward orientation $d\sigma \wedge dx \wedge dy$, and write

$$I_t = \int_{-Y}^{Y} \int_{A}^{B} \int_{-b}^{b} f(t, \sigma, x, y)\, dy\, dx\, d\sigma.$$

The entire remaining outward-face contribution is

$$\begin{aligned} S_\epsilon = & \int_{-\epsilon}^{0} \int_{A}^{B} \int_{-b}^{b} [p(t, Y, x, y) - p(t, -Y, x, y)]\, dy\, dx\, dt \\ & + \int_{-\epsilon}^{0} \int_{-Y}^{Y} \int_{-b}^{b} [r(t, \sigma, B, y) - r(t, \sigma, A, y)]\, dy\, d\sigma\, dt \\ & + \int_{-\epsilon}^{0} \int_{-Y}^{Y} \int_{A}^{B} [s(t, \sigma, x, b) - s(t, \sigma, x, -b)]\, dx\, d\sigma\, dt. \end{aligned} \tag{33}$$

Proposition 7.1 (Complete physical-cap identity)**.** *Suppose the first jet is nonzero on ∂W_- and its interior joint zeros are isolated. Then*

$$I_0 = I_{-\epsilon} - S_\epsilon + Q_-, \qquad Q_- = \sum_{p \in \operatorname{int} W_-} \binom{m_p}{2} \geq 0, \tag{34}$$

where m_p is the spatial multiplicity of the heat zero at p. In particular $I_{-\epsilon} - S_\epsilon$ is a lower bound for the same whole cap I_0.

Proof. Remove small four-dimensional balls about the joint zeros. On the remaining domain $\mathcal{C}$ is the pullback of a closed three-form on S^3. Stokes' theorem therefore identifies the complete outer boundary integral with the sum of the local degrees on small spheres oriented outward from the removed balls, opposite to their induced orientations as holes. The local holomorphic heat-degree calculation assigns a spatial multiplicity m the positive degree $\binom{m}{2}$; this is the local result used in Investigation Note 02, Section 2. Substituting the outward orientations in (32) yields (33) and (34). □

A joint zero on a cap does not meet this proposition's hypotheses. Removing it requires retaining the actual partial sphere and deleted cap piece; no automatic half-integer or one-sided replacement is available. Likewise, shared computational interfaces cancel only when they are the same physical face with opposite orientation. A changed time, radius or population creates a joining contribution. Optional root charts must be supplemented by their full spatial complement. Even a prepared factorization $H = UP$ does not identify its discriminant with the physical root product: for N_0 interior roots,

$$\prod_i H_z(\tau, z_i) = (-1)^{N_0(N_0-1)/2} \operatorname{disc}(P) \prod_i U(\tau, z_i).$$

The nonzero unit remains part of the physical observable.

7.1 Common time and a height graph are different constructions

Along a common complex-time path $\tau = \gamma(s)$, a zero-free spatial wall preserves the complete population, and its prepared coefficients remain holomorphic through internal collisions. Moving the wall does not change its contour moments while the same complete population remains enclosed. This fact does not control the spectrum of a pair matrix on a complex slice. If $R = (I + aL)^{-1}$ and $q = \mathrm{Tr}(I - R)$, then

$$\frac{dq}{ds} = \mathrm{Tr}\big[R(a'L + a\gamma' L_\tau)R\big] .$$

A matrix-norm estimate requires a lower singular-value bound for $I + aL$; its two derivative terms should remain coupled. Distinct roots alone do not guarantee this resolvent condition. For example, $H = z^2 - 1 - 2\tau$ has $q = 2\tau/(1+4\tau)$, whose pole at $-1/4$ is not a root collision. The coefficient/forest method elsewhere in this paper provides a different scalar denominator test, with its own stated scope.

A height graph instead samples $F(z) = H(h(\Re z) + i\sigma, z)$, with real C^1 function h. It is generally nonholomorphic:

$$F_z = H_z - \tfrac{1}{2}h' H_{zz}, \qquad F_{\bar z} = -\tfrac{1}{2}h' H_{zz}, \qquad \frac{dF}{F} = \frac{H_z}{H}\,dz - h'\frac{H_{zz}}{H}\,dx. \tag{35}$$

The resulting winding is a topological degree, rather than an unsigned holomorphic zero count. At a graph zero with $H_z \neq 0$, its Jacobian is

$$J_F = |H_z|^2 \Re\!\left(1 - h'\frac{H_{zz}}{H_z}\right).$$

Positive orientation therefore needs a separate transversality condition. For isolated transverse graph zeros, weighted counting also has the bulk correction

$$\sum_{F(\zeta)=0} \mathrm{ind}_\zeta(F)\zeta^k = \frac{1}{2\pi i}\oint_{\partial\Omega} z^k \frac{dF}{F} + \frac{k}{\pi}\int_\Omega z^{k-1}\frac{F_{\bar z}}{F}\,dA, \qquad k \geq 1.$$

This follows by Stokes' theorem on the punctured domain; the integrand is locally integrable at a transverse zero. A negative graph $t = -\epsilon(x)$ also changes the physical cap density to $f + \epsilon'(x)r$. Neither correction is optional. These distinctions and finite graph comparison bounds are proved in Investigation Note 08, Sections 6–7.

7.2 What the comparison has established

The direct estimator receives the same complete source representation, precision, signed correlations, wall data and any acquired factorization as the deformed estimator. A common analytic endpoint identity does not by itself make a detour cheaper. Under a uniform residual allowance e, a path pays its length times e; a longer path can improve that budget only if its actual source or conditioning bounds improve enough. This is a scoped comparison, not an exclusion of useful contour deformation.

The finite canonical test uses the original box

$$[26, 46] + i[-2, 2], \qquad \tau \in [-10^{-4}, 10^{-4}] + i[-10^{-4}, 10^{-4}].$$

An interval certificate supplies $E > 1.1178 \cdot 10^{-13}$ throughout the whole box, with no deleted root disks. Thus $Q_- = 0$ and both estimators know the exact equality in (34). A single bounded range-quadrature pilot, using 512 spatial leaves and all the faces in (33), nevertheless gives only

$$I_0 = 2.0247388041 \cdot 10^{-5} \ \pm 1.42 \cdot 10^{-4}, \qquad I_{-\epsilon} - S_\epsilon = 2.0247432516 \cdot 10^{-5} \ \pm 1.42 \cdot 10^{-4}.$$

These are certified enclosing intervals, not accurate midpoint estimates. Both lower endpoints are negative. The raw deformed width is slightly larger, approximately 0.000283934 versus 0.000283522, and it requires additional face evaluations. The close σ faces were subtracted algebraically before interval bounds, retaining their factor $2iY$. The total side-face interval has radius below $2.05 \cdot 10^{-7}$; spatial range overestimation of the caps dominates the uncertainty. Investigation Note 15 and its replay give the complete budgets and scope.

Accordingly this finite experiment pays the source, regularity and full boundary-accounting obligations, but demonstrates neither a positive complete cap lower bound nor a contour advantage. It does not justify a larger-height campaign. A further numerical comparison, if undertaken, should test higher-order correlated spatial integration on the same observable and at a fixed total budget, with the same improvement made available to the direct baseline.

8 Research priorities after the finite comparison

The strongest completed structural result is the complete-source coefficient equation together with its commuting finite propagator. The forest polynomial turns it into a quantitative scalar-filter interface. The corrected Riemann–Siegel reference is the most concrete theta-specific input now available on the microscopic clock. These results fit together, but their composition still contains genuine acquisition requirements.

Result	Assumption or remaining input
Forest quotient	Polynomial algebra; a paid denominator margin for an enclosure.
Coefficient transport	Complete population and a common nonzero wall; integrated complete-source moments and errors.
Positive-time multiplicity transfer	Complete real adjacent-root motion; RH in the stated canonical global application.
Microscopic corrected reference	Imported real-time source theorem, fixed truncation chart; whole-wall margin still unacquired at growing height.
Finite theta certificate	Complete-source interval errors on the specified box; no extension to unbounded height.

8.1 A bounded next acquisition with a decisive question

A useful next test would fix one genuinely high-height chart before examining its zeros. For example, put $\mathcal{L} = \log(x/(4\pi))$ and prescribe $t = \mathcal{L}^{-3}$, with spatial scales proportional to $\mathcal{L}^{-1}$. With natural unfolding $\nu \sim \mathcal{L}/(4\pi)$, this gives $4t\nu^2 \to 0$. The choice is a proposed experimental design, not a theorem about its resulting roots. A fixed Riemann–Siegel cutoff, or explicitly paid cutoff changes, is required.

The test should acquire the corrected normalized reference on every side of the declared wall, its absolute source error, the complete population count, coefficient errors, and the coupled forest denominator/residual margin. It should report the resulting filter bound alongside an equally informed direct fixed-time calculation. The decisive question is which term fails: the wall margin, the source-to-coefficient conversion, degree growth, the forest denominator, or the reference filter itself. One successful chart is a validation of the acquisition method, not a limiting law. A broad sweep is unjustified unless it tests a specified uniform estimate or cost model.

8.2 The arithmetic estimate cannot be replaced by conditioning

The long-term target is (4): a vanishing normalized filter and vanishing omitted mass on a family of complete populations. Better source errors make a reference trustworthy; they do not make its filter small. Even excellent whole-wall conditioning can enclose an interior multiple zero. An independent arithmetic estimate must ultimately control the relevant clustering of the complete reference, or the unchanged field itself, on the prescribed joint height/time scale.

An averaged one-sided small-value estimate for the complete normalized field and derivative is a possible direction, provided a proved consumer connects it to the needed root counts and every interpolation, exceptional-set and coverage loss is paid. This is a research suggestion, not an estimate supplied here. Generic real-root heat dynamics, a mean zero count, or a statistical model cannot stand in for the missing canonical input.

8.3 Conditional roles for backward transport and curved paths

A further backward calculation is justified when it can acquire the full correlated residual in (13), including the explicit reference heat defect, and improve the final certified error or cost relative to direct acquisition from the same information. The finite propagator alone does not establish that improvement. The scalar late-time implementation with separately bounded algebraic errors has a different, unfavorable amplification budget; its failure does not exclude a joint source estimate.

Complex time should be pursued only with a proposed complete-source error bound that improves a specified path budget. Holomorphic continuation of the reference is insufficient because the current imported field theorem is for real time. A cap calculation must additionally retain every connector and complement. The completed low-height range-quadrature pilot identifies a spatial integration bottleneck, so further source precision or geometrical rephrasing would not address its dominant error. Higher-order correlated integration could be a bounded numerical study, with the same improvement available to the direct baseline; it is a lower priority than the microscopic arithmetic and conditioning questions.

The present stopping point is therefore a finite method with an explicit source interface and an honest comparison, not an almost-complete proof of simplicity. Density-one simplicity and every-zero simplicity remain distinct targets; the latter cannot discard a sparse family of doubles. An RH-free application must also establish endpoint reality independently.

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Research Notes 07–15, their reviews and small execution records are maintained in the parent investigation directory. The earlier analytical controls include 1,333 counted exact-rational assertions, with separate floating diagnostics. The complete-source certificate uses Arb interval arithmetic through python-flint, as described in its section; numerical checks support the stated finite claims only. Sources, receipts and provenance are kept separately so the mathematical assertions do not depend on a file hash.

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